AUTOMATION CASE STUDY

Pre-renewal retention loop

A local n8n demo that asks earlier, reads the reply, and brings the right issue to a property manager before the formal renewal conversation.

event_type	channel	direction	lease_id	tenant_name
DRAFTED	SYSTEM	INTERNAL	cedar-row-9-2026-11-11	Avery Brooks
SENT	EMAIL	OUTBOUND	cedar-row-9-2026-11-11	Avery Brooks
REPLY_RECEIVED	EMAIL	INBOUND	cedar-row-9-2026-11-11	Avery Brooks
ASSIST_SENT	EMAIL	OUTBOUND	cedar-row-9-2026-11-11	Avery Brooks
SENT	EMAIL	OUTBOUND	cypress-4-2026-11-07	José Núñez
REPLY_RECEIVED	EMAIL	INBOUND	cypress-4-2026-11-07	José Núñez
RESOLVED	SYSTEM	INTERNAL	cypress-4-2026-11-07	José Núñez

OUTREACH LOG / actual Airtable demo output

The problem: lease dates are visible, but unresolved resident concerns can remain hidden until a move-out decision is already forming.

The build: date-based check-ins, individual approval in Telegram, Gmail reply matching, issue routing, and a live Airtable record of what happened.

SYSTEM DESIGN

One connected operating loop

n8n coordinates five workflows. Airtable holds demo state; Telegram is the manager's action surface; Gmail carries resident messages.

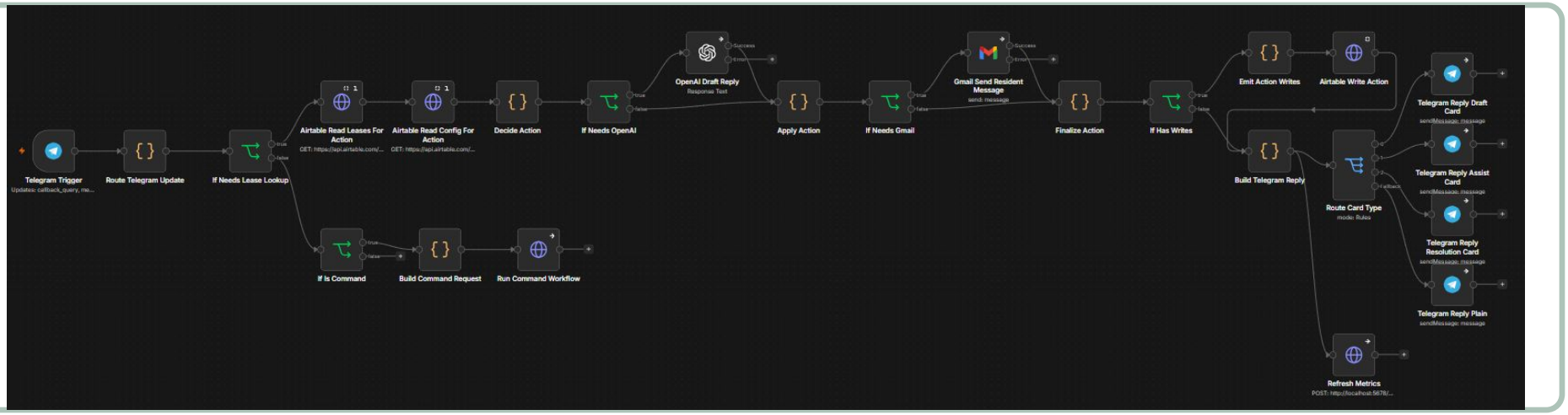
01	Airtable	Synthetic lease roster, drafts, reply history, issue state, and metrics
02	n8n + OpenAI	Date-based selection, constrained drafting, classification, routing, and sweeps
03	Telegram	One bot receives approvals, edits, replies, resolution actions, and alerts
04	Gmail	A single send path delivers only messages approved for that resident

The cadence

120 days: watchlist. 90 days: manager-approved check-in. 60 days: a nudge only if unanswered, again with approval. 30 days: internal handoff, with no resident email.

Every manager action enters here

The bot has one Telegram trigger. This workflow interprets a button or command, fetches current state, and runs the approved action.



FULL CANVAS / 25 nodes visible

- Trigger: an inline button, slash command, or text edit from the manager's Telegram chat.
- Decision: read the resident and configuration, then decide whether to draft, send, skip, resolve, or show the next card.
- Output: one Gmail message only after a human tap; Airtable and Telegram receive the updated state.

Human gate at the last possible moment

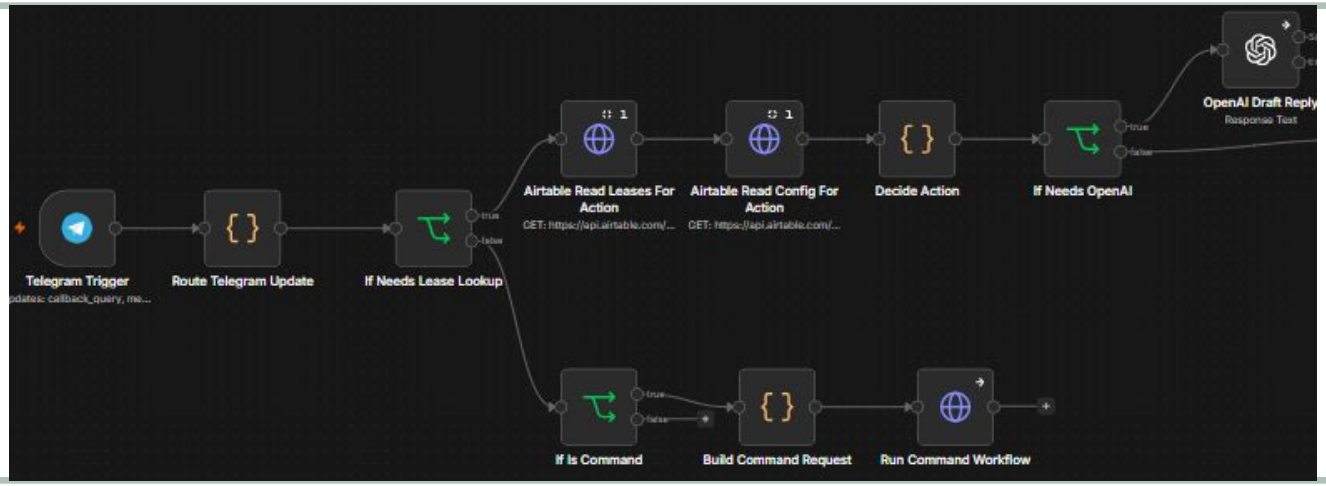
The current Airtable wording is checked again when Approve is tapped. A second tap cannot create a second email.

WORKFLOW 01 / ENLARGED

From button to send decision

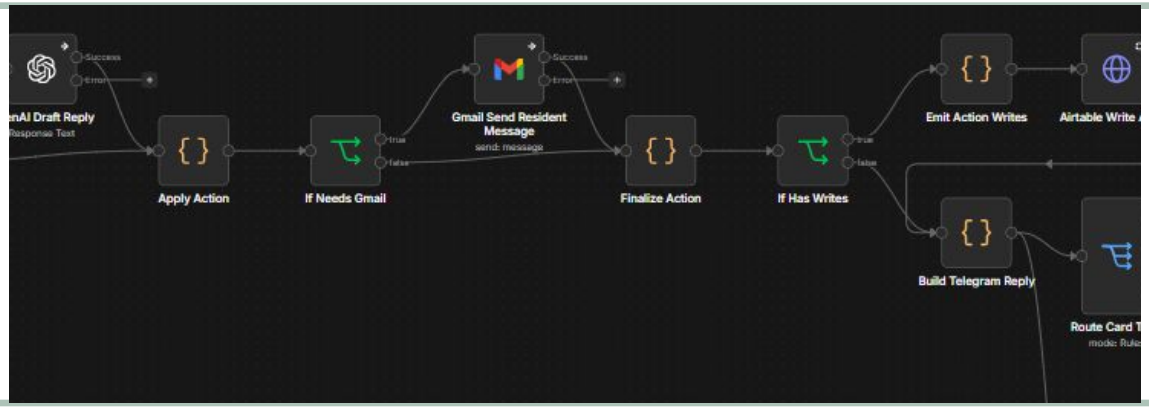
The two crops follow the same canvas from Telegram input through the resident send gate.

A / ACTION ENTRY



The router separates commands from resident-specific actions. It reads the lease and settings before applying any button press.

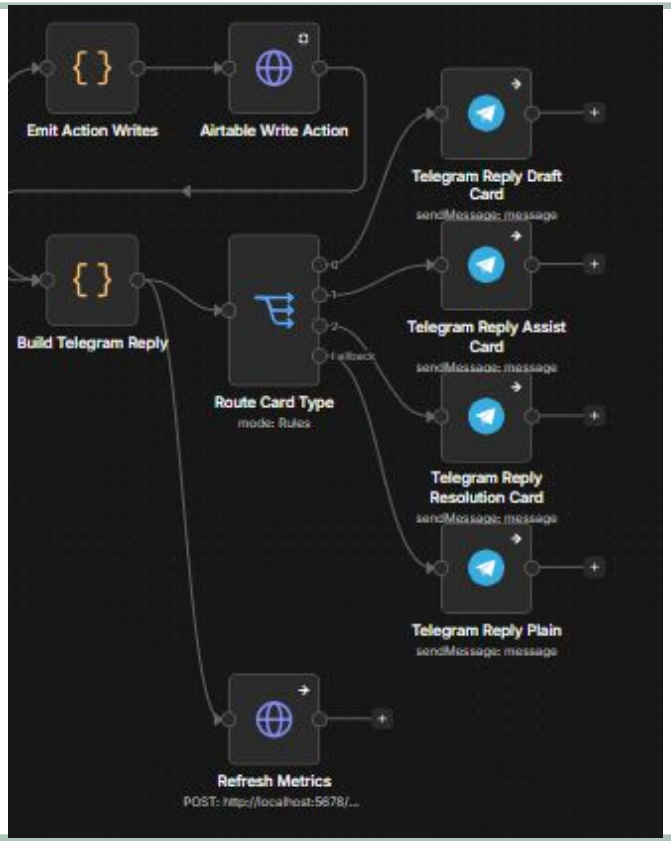
B / APPROVAL AND GMAIL GATE



Reply assistance can be drafted here, but the Gmail node is reached only when the manager approves an individual message.

Write state, then answer the manager

After the decision, the workflow records its effect and returns a card that matches the next available action.



C / AIRTABLE WRITES, METRICS REFRESH, AND TELEGRAM CARD ROUTING

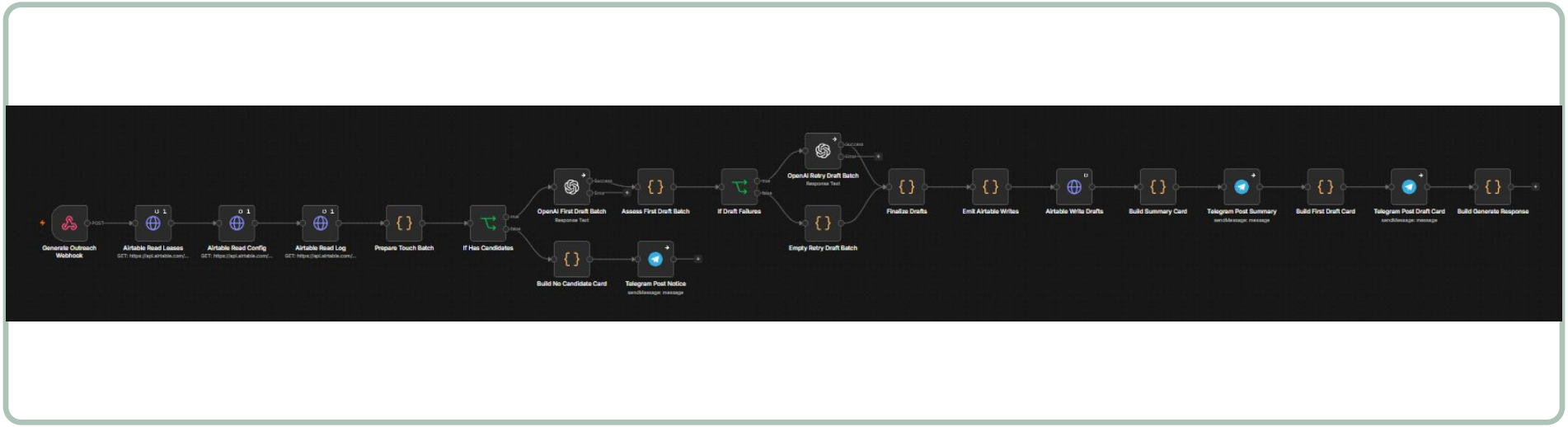
Why the outputs split

Different Telegram card layouts carry different inline buttons. The workflow also refreshes metrics after an action, so Airtable reflects the latest demo state.

WORKFLOW 02 / OUTREACH

Prepare the check-in queue

A webhook starts a dated outreach run. This workflow creates reviewable drafts and approval cards; it does not send a resident email.



FULL CANVAS / 21 nodes visible

- Reads leases, settings, and the outreach log to select eligible 90-day or unanswered 60-day residents.
- OpenAI drafts in a batch; fixed wording rules assess each result and retry a failed draft once.
- Writes approved-for-review drafts to Airtable and posts a summary plus the first Telegram approval card.

Verified generation

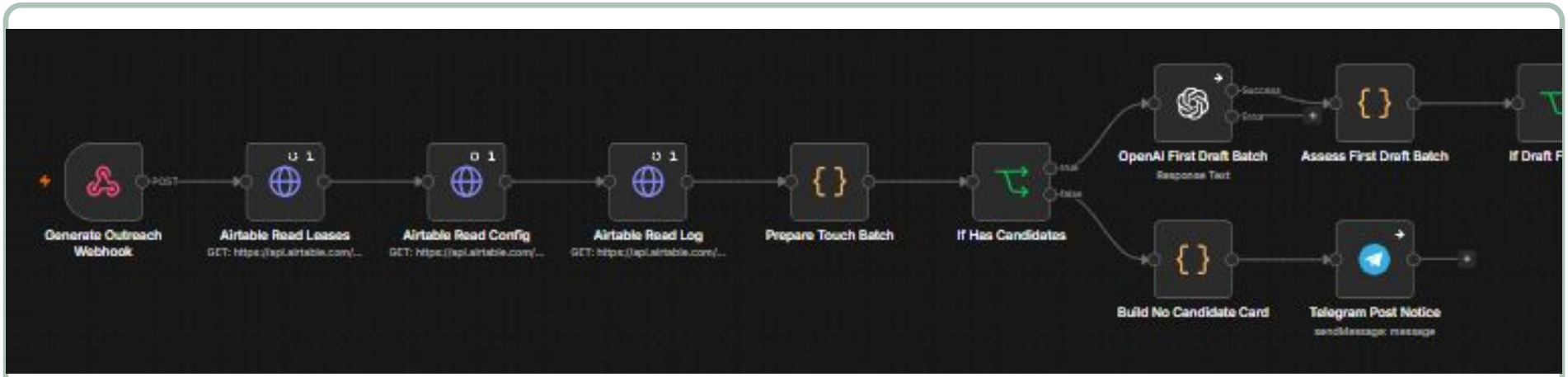
In the recorded synthetic run, all 14 of 14 90-day drafts passed the fixed guardrails before review.

WORKFLOW 02 / ENLARGED

Selection, drafting, and review

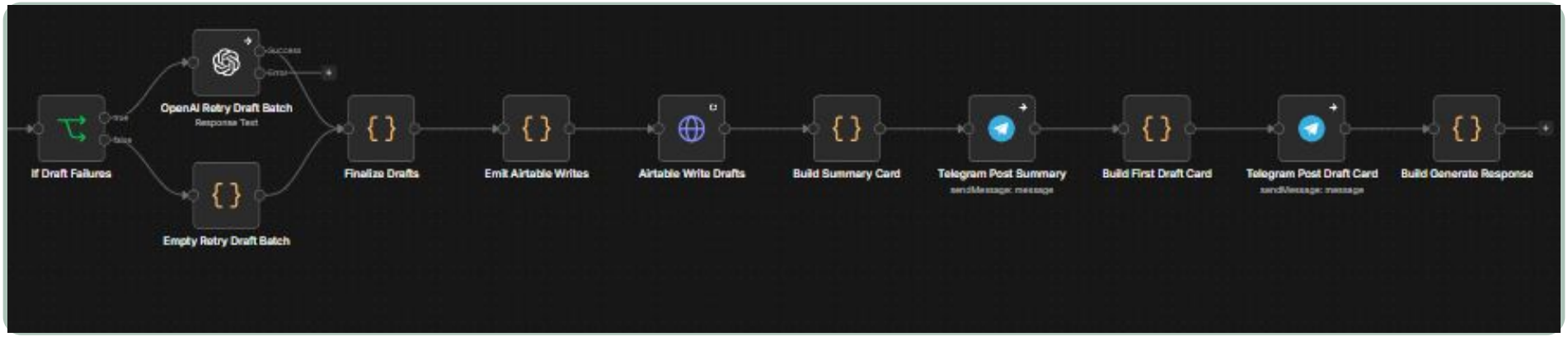
The first crop is the deterministic candidate filter; the second shows model drafting, retry, writes, and manager cards.

A / READ, SELECT, AND BRANCH



Cohort membership comes only from the lease end date. Prior sends and opt-outs are checked before a resident enters the batch.

B / DRAFT, CHECK, WRITE, NOTIFY

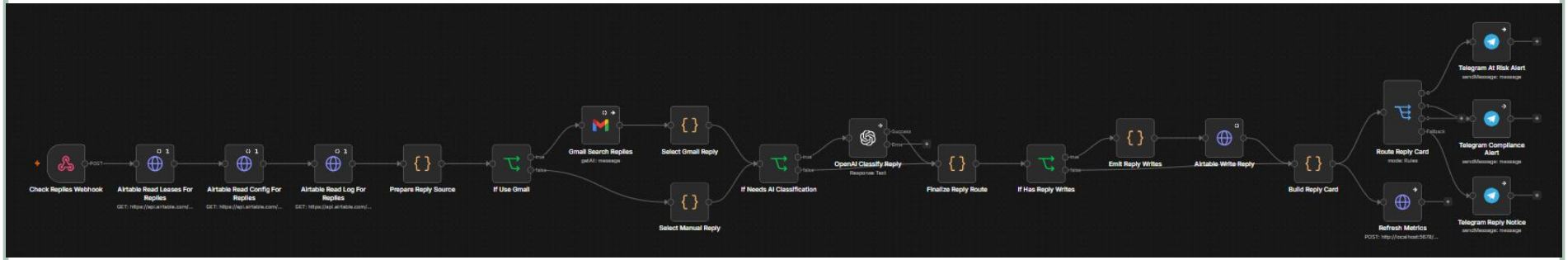


A draft that fails twice is left for manager wording. A usable draft becomes an Airtable row and a Telegram review card.

WORKFLOW 03 / REPLIES

Turn a reply into a clear next action

A reply check reads Gmail or a controlled test input, identifies the resident without asking the model, and routes the meaning.



FULL CANVAS / 21 nodes visible

- Matches a reply by its [R-xxxx] subject token, then by exact sender address; unmatched mail stays unassigned.
- Fixed rules catch sensitive categories first. Structured OpenAI classification handles the remaining replies.
- Writes the reply and issue state to Airtable, refreshes metrics, and alerts the manager when judgment is needed.

The signal it caught

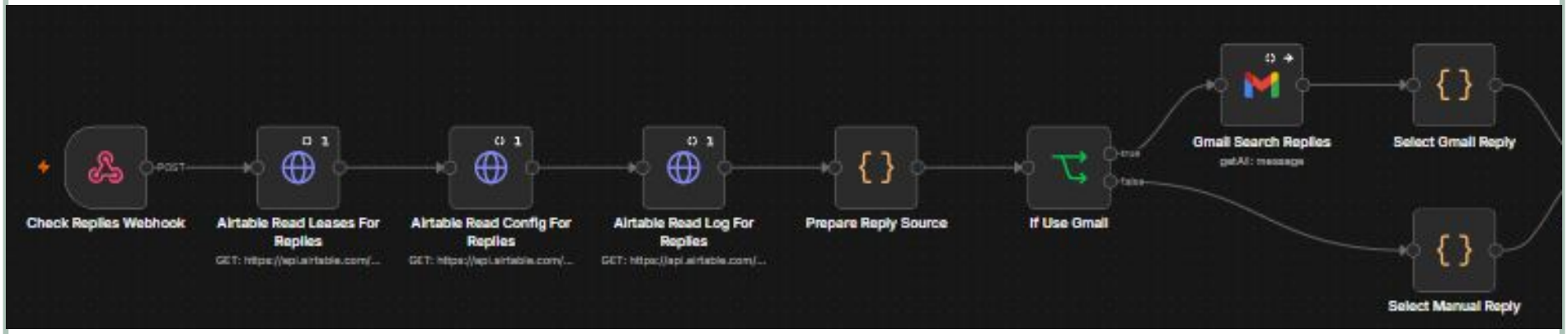
A positive-sounding reply that also described a long-standing broken extractor fan was recorded as a high-urgency maintenance grievance.

WORKFLOW 03 / ENLARGED

Match first. Interpret second.

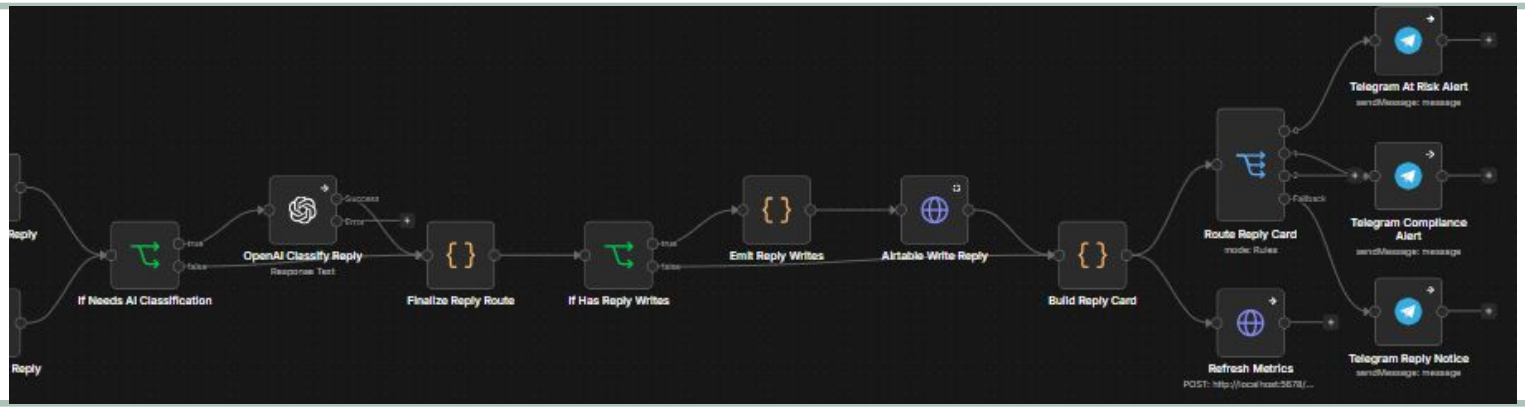
These crops make the separation between identity matching and AI interpretation visible.

A / FIND AND MATCH THE MESSAGE



The workflow reads the mailbox and the existing log, then skips messages already handled or predating the demo's first send.

B / CLASSIFY, WRITE, AND ALERT



A clean high-confidence positive can close green. Grievances, questions, uncertainty, opt-outs, and sensitive flags go to a person.

WORKFLOW 04 / OPERATIONS

Keep the roster and scoreboard current

A scheduled sweep or internal webhook recomputes timing, writes metrics, and posts a digest when requested. Errors have their own alert path.



FULL CANVAS / 18 nodes visible

- The daily trigger rereads leases, configuration, outreach history, and the current metrics row.
- The sweep recomputes cohorts and unresolved non-response, then writes changed rows in batches.
- A separate error trigger tells the manager if an automation run fails rather than failing quietly.

The scoreboard's meaning

Exposure surfaced and exposure recovered apply a configurable demo turnover cost to issue state. They are not revenue or savings claims.

WORKFLOW 04 / ENLARGED

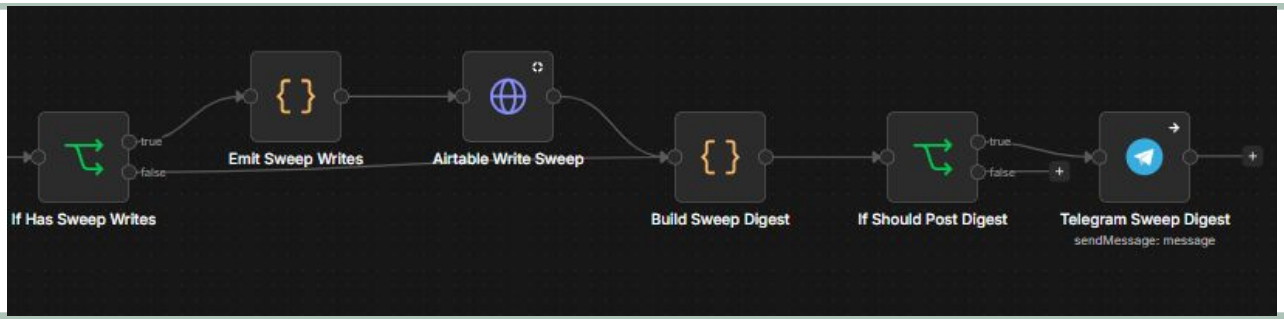
Sweep, digest, and visible failure

The error-alert branch is independent of the scheduled metrics path.

A / SWEEP INPUT AND RECOMPUTE



B / BATCH WRITES AND TELEGRAM DIGEST



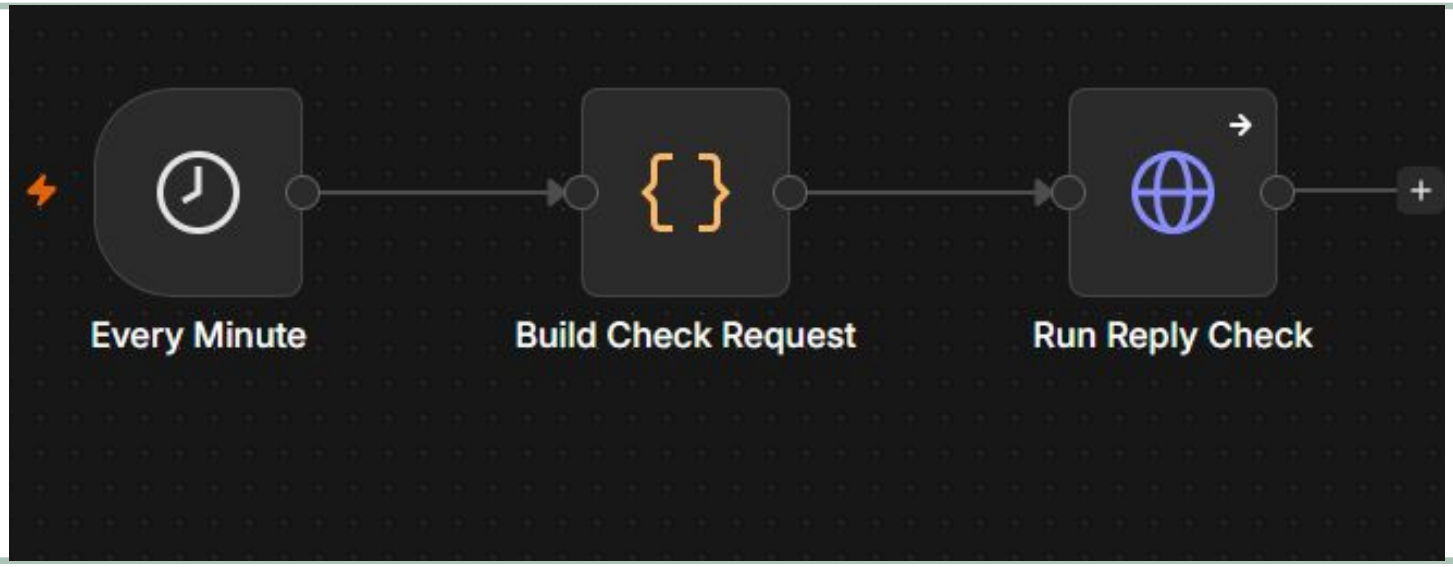
C / WORKFLOW ERROR ALERT



WORKFLOW 05 / AUTOMATIC REPLY CHECK

The quiet minute-by-minute poll

A short scheduled workflow invokes the reply check. It stays silent when the mailbox has nothing new.



FULL CANVAS / ALL 3 NODES VISIBLE

- Trigger: a one-minute n8n schedule.
- Key steps: build a check request and call the reply-routing webhook.
- Output: the reply workflow takes over only when it finds a new matching message; this workflow contains no resident send node.

How the five workflows connect

The poller wakes reply routing. Telegram handles human actions. Outreach prepares drafts. The operations workflow updates cohorts and metrics after each meaningful event.

VERIFIED DEMO OUTPUT

Synthetic roster, visible decisions

These are actual Airtable screens from a recorded demo run. All residents shown are synthetic fixture identities.

A / DATE COHORTS AND DATA-QUALITY FLAGS

days_to_expiry	cohort	tenure_months	data_quality	needs_attention
90	90	18		
90	90	28		
88	90	3	TENURE_UNDER_6M	✓
128	120	21		
48	60	28		
110	120	49		

B / REVIEW QUEUE AND RESIDENT STATE

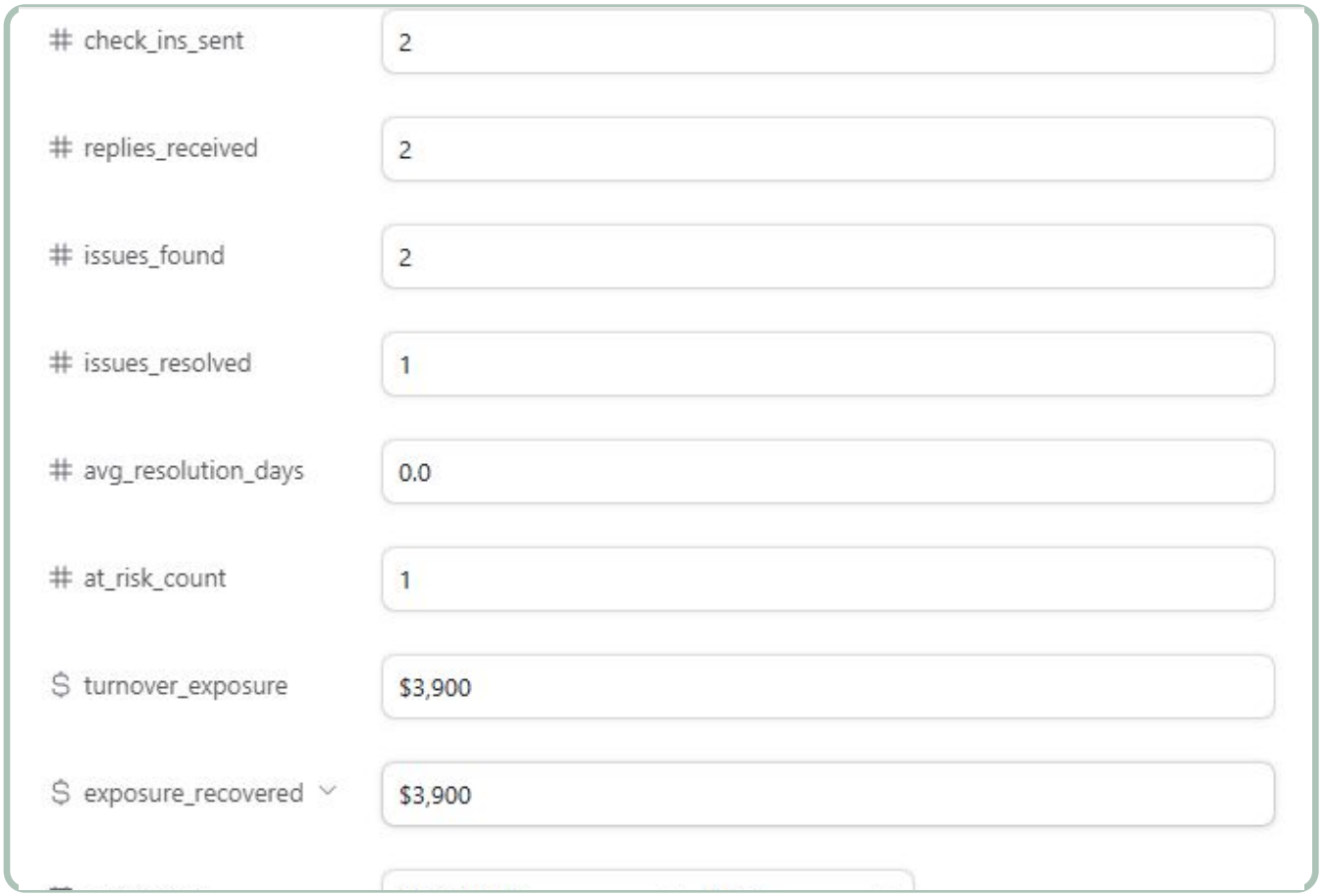
	lease_id	status	touch_stage	draft_subject
1	cedar-row-9-2026-11-11	AT_RISK	CHECK_IN_90	Checking in about Cedar R...
2	cypress-4-2026-11-07	RESOLVED	CHECK_IN_90	A quick home check-in [R-...
3	ash-court-2d-2026-12-01	AWAITING_APPROVAL	CHECK_IN_90	Checking in on your home ...
4	linden-house-14-2026-11-19	AWAITING_APPROVAL	CHECK_IN_90	A note about Linden House...
5	canyon-ridge-15-2026-12-...	AWAITING_APPROVAL	CHECK_IN_90	A check-in about Canyon R...
6	elm-terrace-6-2026-11-23	AWAITING_APPROVAL	CHECK_IN_90	Checking in after your time...

The roster contains 40 synthetic residents. The run grouped 14 into the 90-day window and kept 8 flagged rows visible for attention.

VERIFIED DEMO OUTPUT

The scoreboard records the loop

An actual Airtable metrics record from a synthetic demo run, shown at readable size.



METRICS / recorded synthetic snapshot

14 in the window. 2 check-ins sent. 2 replies received. 2 issues found. 1 issue resolved. One issue remained open in this snapshot.

The $3,900 exposure figures use a configurable demo assumption. They represent issue state, not earned savings.

VERIFIED DEMO OUTPUT

From message to recorded issue

Airtable retains the event sequence and the routed issue category for each synthetic reply.

A / OUTREACH EVENT LOG

event_type	channel	direction	lease_id	tenant_name
DRAFTED	SYSTEM	INTERNAL	cedar-row-9-2026-11-11	Avery Brooks
SENT	EMAIL	OUTBOUND	cedar-row-9-2026-11-11	Avery Brooks
REPLY_RECEIVED	EMAIL	INBOUND	cedar-row-9-2026-11-11	Avery Brooks
ASSIST_SENT	EMAIL	OUTBOUND	cedar-row-9-2026-11-11	Avery Brooks
SENT	EMAIL	OUTBOUND	cypress-4-2026-11-07	José Núñez
REPLY_RECEIVED	EMAIL	INBOUND	cypress-4-2026-11-07	José Núñez
RESOLVED	SYSTEM	INTERNAL	cypress-4-2026-11-07	José Núñez

The log shows drafting, individual sends, replies, an approved assist, and a resolution event.

B / REPLY CLASSIFICATION IN THE RESIDENT RECORD

intent	grievance_category	urgency	compliance_flag	confidence
GRIEVANCE	MAINTENANCE	HIGH	NONE	HIGH
GRIEVANCE	PROPERTY_CONDITION	HIGH	ACCOMMODATION_RE...	HIGH

The captured records include a high-urgency maintenance grievance and an accommodation request flagged for human review.